# Term Project (account for 20% of total score)

- Assigned as individual project
- **Descriptions**
  - For the project, you are asked to implement CUDA programs to process large data sets
  - Step 1: Pick your favorite problem as well as the data set that is publicly available
    - You are not allowed to choose the problems in the lab assignments
  - Step 2: Write the best possible sequential solution. Also argue why it is the best possible
  - Step 3: Design your own parallel solution to solve the problem on the large data set. Argue why one would make senses about the solution (i.e., the solution should be good)
  - Step 4: Implementing the sequential solution in terms of CPU code
  - Step 5: Implementing the parallel solution in terms of CPU-GPU code
- **Requirement**
  - Must demonstrate that your parallel solution outperforms the sequential counterpart on large data

# Scoring Criteria and Due Time

- Total score 20%
  - Validity of the entire project (e.g., the chosen problem and data set) 5%
  - Creativity 5% (e.g., technique to achieve parallel efficiency )
  - Correctness of program codes 5%
  - Presentation 5%

- Submission: The GitHub link to your project
  - The project should have
    - A folder containing all the required files
    - A document file containing info about how to run the project
    - The presentation file
  - Due date: Wed 4 Nov (11.59 PM)

- Presentation: Each individual needs to present his project (presentation time is about 15 mins including Q&A)
  - Presentation date: Th 5 Nov (13.00-15.00; Room TBD)